

Getting Started with GitHub for MART 145

GitHub is a web-based platform for version control and collaboration on software projects. It allows developers to store their code in "repositories," track changes over time, and work together on larger projects by suggesting, reviewing, and merging code changes. GitHub can host and manage almost any type of digital document. While it was originally used by software developers for version control, GitHub's functionality has expanded to allow the creation of a profile, serve as a social network for developers, and can be used to host a website.

How will we use GitHub in MART 145?

- File storage. You will set up a repository (which is like a folder) and upload your website files that you create for the class. This serves as a backup, and a way to share files with your instructor for grading.
- Web Hosting. GitHub displays front-end websites (HTML, CSS, JavaScript) correctly in the browser, making it a free and easy way to make your website available to the world. GitHub Pages allows you to web-enable a repository. This means free web hosting!

Overview and Terminology

Git for Version Control

GitHub is actually a bundle of software tools. Git is the underlying open-source version control system that tracks every change to a project's files, allowing users to revert to older versions if needed. You can download a desktop application that allows you to use Git on the command line. This is informational and you do NOT need to do any of this now.

Repositories

A repository is a folder where you will store your code and project files. You should create repository for the class and set it's visibility to Private. You can create additional repositories for your classes and projects. A repository serves as cloud storage, and a backup in case anything goes wrong.

Collaboration

You do NOT need to use GitHub for collaboration in this class. However, you may come across documentation that describes pull requests, branching, and approve changes. These are all part of the collaboration functionality.

Web Hosting

GitHub Pages is the product used for web hosting. Pages requires you to create a repository for your website, and it must conform to a naming scheme. This repository will automatically be web-enabled.

Instructions

Set up an account

This is a fairly straight-forward process. Browse to <https://github.com/>, click Sign Up and follow the instructions. Read the GitHub instructions if you need more assistance:

<https://docs.github.com/en/get-started/start-your-journey/creating-an-account-on-github>

Create a repository

Log on to GitHub. You will typically be directed to the Dashboard after logging in successfully. The repository menu will be on the left-hand side of the screen. Click the green New button. This will bring you to the screen where you can name your repository and add additional information such as a description and visibility settings. Any repositories used to store class files should be set to Private visibility. The GitHub instructions are here:

<https://docs.github.com/en/repositories/creating-and-managing-repositories/creating-a-new-repository>

Create your GitHub profile

Log on to GitHub. There will be a small icon on the upper right corner of the screen. This allows you to access your account information. Click on the icon, and select Your Profile. Add any information about yourself that you would like to share, and save your changes.

Public repositories will appear on your profile page.

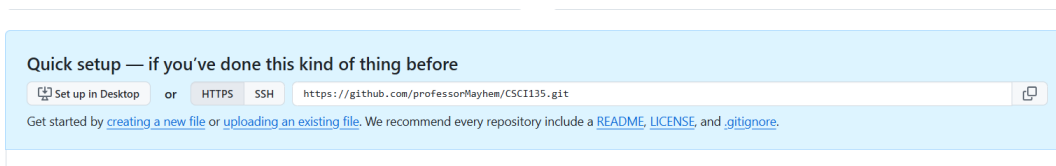
Melissa's profile: <https://github.com/professorMayhem>

Explore and Navigate

Click on the various menus and icons in GitHub, and get a feel for its interface. Be sure you are comfortable with your public profile, and confident creating repositories. Remember, there are many features that we will NOT use in this class.

Storing Files

Practice adding files to a repository. Click on the Repositories button at the top of your public profile. Click on the repository where you want to store your files. If you have never uploaded a file, you will see the following:



Click on uploading an existing file in the tiny blue text. After you upload files, there will be an “Add File” button near the top of the screen.